# Raghav Khandelwal

**Frontend / React Developer | Next.js | PWA**

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## Professional Summary

Frontend-focused Software Engineer with **3+ years of experience** building scalable, high-performance web applications using **React, TypeScript, and Next.js**. Delivered production systems used by **1M+ users PAN India**, including platforms for **IIT Bombay, CSIR, and Government healthcare initiatives**. Strong expertise in **PWA development, offline-first architectures, JWT authentication, performance optimization, and API integration**.

## Technical Skills

### Frontend

React.js, Next.js (App Router, Server Components), TypeScript, JavaScript (ES6+), HTML5, CSS3, Tailwind CSS, Bootstrap

### State Management & UI

Redux Toolkit, Context API, Reusable Component Architecture, Form Validation, Responsive Design

### PWA & Performance

Progressive Web Apps (PWA), Service Workers, Offline-first Applications, Lazy Loading, Code Splitting, Client-side Caching, Pagination

### Backend Integration

REST API Integration, JWT Authentication, Role-based Access, Node.js & Express (working knowledge), Python/Django APIs

### Tools & DevOps

Git, GitHub, GitLab, CI/CD Pipelines, Docker, Postman, VS Code, Figma

## Professional Experience

### Frontend Developer / Software Engineer – Virtual Labs, IIT Bombay

*January 2023 – Present*

- Designed and developed UI for **180+ interactive lab simulators** using React and a self-developed TypeScript-based framework.
- Built scalable, responsive interfaces used across **20+ Nodal Centers**.
- Integrated backend APIs for authentication, validation, and workflows.
- Implemented **JWT-based authentication** and role-based access.
- Improved performance using reusable components and lazy loading.
- Worked in **Agile (Scrum)** environment with CI/CD pipelines.

## Key Projects

### Central Outreach Portal – IIT Bombay

- National-level platform serving **1.2+ million annual users**.
- Built reusable forms and improved load performance by **~40%**.
- URL: https://iitb.vlabs.co.in/outreachportal/

### Jigyasa Platform – CSIR

- Used by **120,000–140,000 users PAN India**.
- JWT authentication, PWA with offline support, mobile installable.
- URL: https://jigyasa.csir.in

### Digital Health Platform – Ayushman Bharat

- PWA-based medical data collection app for health workers.
- Offline-first design, Redux state management, Dockerized dev.

## Certifications & Courses

- Artificial Intelligence & Machine Learning – IIT Guwahati
- Machine Learning A–Z (Python) – Udemy
- Algorithms & Data Structures in Python – Udemy
- Design & Verification Using Verilog – Entuple Technologies
- Software Engineering Skill Development – Cutshort

## Education

**B.Tech – Electronics & Communication Engineering**
Vellore Institute of Technology, Bhopal
GPA: 8.56 / 10